

Test 4 - Mixed Realistic 24-Hour Operation Test

Purpose

The purpose of this test is to verify that the complete EMS behaves correctly under realistic operating conditions, using real API price data, the adjusted load profile, a realistic PV/lamp profile, and naturally evolving virtual SoC values.

This test should confirm that the EMS follows the expected scenario logic at each timestep:

Condition	Expected scenario
LOW price + no PV	S1
LOW price + PV + battery not full	S2
LOW price + PV + battery full	S3
HIGH price + PV available	S4
HIGH price + no PV + battery usable	S5
HIGH price + no PV + battery unusable + PEM SoC > 10% + load < 40 mW	S6
Otherwise	S1

The main purpose is not necessarily to reach every scenario, but to verify that the selected scenario matches the system state throughout a realistic test.

Initial setup

Using Scripts:

AppLab: ems_base_estimation_logger_v5

Arduino IDE: ControllableLoad_V4.1_triggerAndData

Item	Value
Test duration	24 minutes
Simulated duration	24 hours
Price data	Real API day with mixed LOW/HIGH prices

Load profile	Active, 15–75 mW
Initial Battery SoC	90%
Initial PEM SoC	50%
PV/lamp at start	Off
CSV logging	Enabled

Demo and EMS control

Start the EMS system first. Then start the demo to reset the 96-slot timeline, begin CSV logging and activate the variable load trigger. Raw logging can run independently of the demo.

EMS: started

Bridge: OK

Reset

Start EMS system

Stop EMS system

Start demo + logging

Stop demo

Auto mode

Manual mode

Start raw log

Raw log

Stopped

Raw log file

-

Raw samples

0

Last raw update

-

Price test date

09/05/2026

Manual electricity price

0.99729

Hydrogen estimation (%)

50.0

Virtual battery charge state (%)

90.0

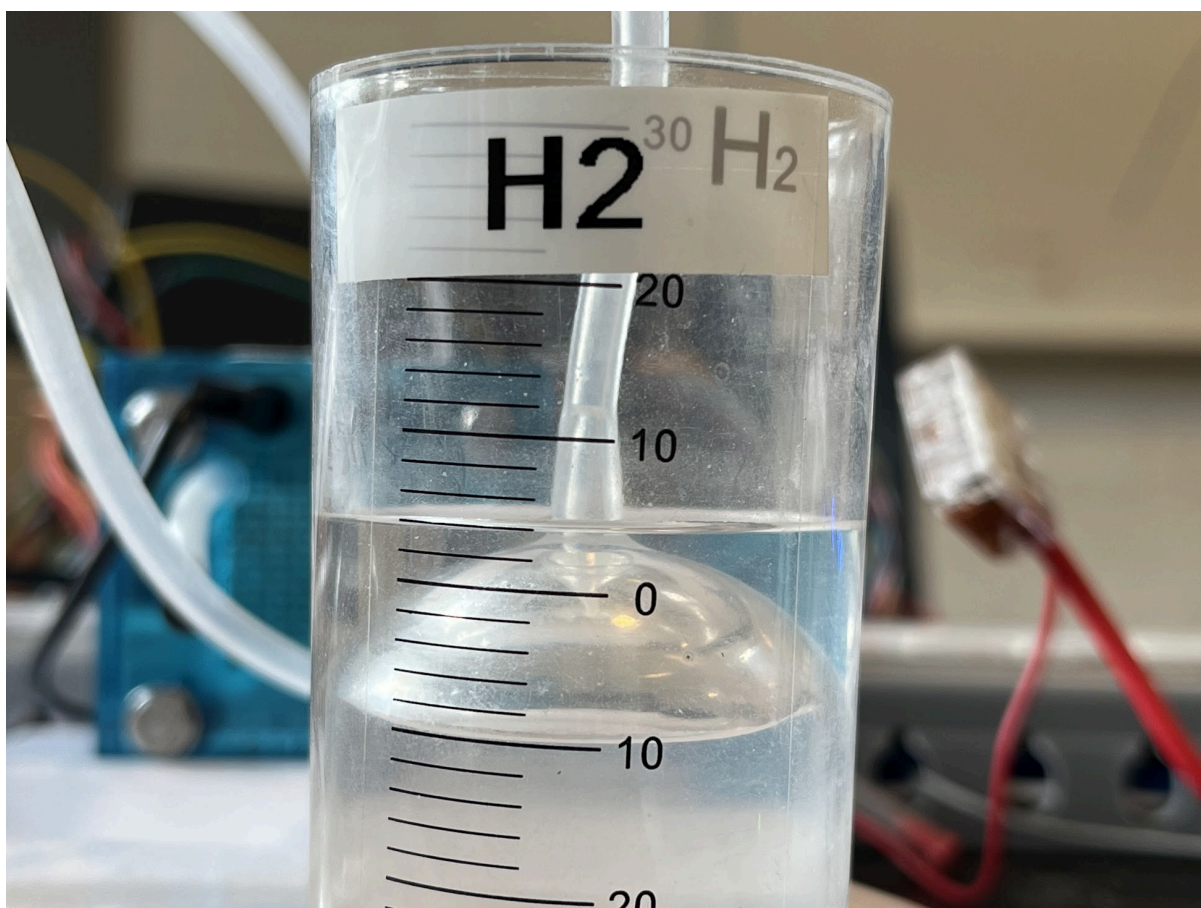
Load prices for date

Apply manual price

Initiate H2 estimation manually

Apply virtual battery state

Storage states		INA sensor measurements	
Current battery voltage	3.693 V	PV voltage	3.229 V
Initial SoC lookup voltage	3.692 V	Battery voltage	3.693 V
Lookup-estimated real battery SoC	4.7 %	PEM voltage	1.285 V
Integrated real battery SoC	4.6 %	Load voltage	5.128 V
Integrated real battery charge	92.959 mAh / 2000.0 mAh	PV current	0.15 mA
Virtual battery SoC	90.0 %	Battery current	-0.03 mA
Virtual battery capacity	10.00 mAh	PEM current	-0.04 mA
Virtual battery charge	9.000 mAh	Load current	-0.01 mA
Virtual battery state	control_full	PV power	0.48 mW
Hydrogen volume	11.60 mL	Battery power	-0.09 mW
Usable hydrogen	4.00 mL	PEM power	-0.05 mW
Hydrogen estimation	50.0 %	Load power	-0.04 mW
Hydrogen mode	idle	Load trigger	LOW
		Arduino mode	S1 Grid -> Load



Hydrogen Niveau inden test (50% = 4.00 mL) (formentlig målt forkert ved start)

Test procedure

1. Load the selected real API price day with mixed LOW/HIGH prices.
2. Set Battery virtual SoC to 90%.
3. Set PEM virtual SoC to 50%.
4. Start the load profile.
5. Start with the lamp off.
6. Start CSV logging.
7. Start the EMS test.
8. Follow the planned PV/lamp profile during the 24-minute run.
9. Observe scenario changes caused by price, PV availability, SoC, and load.
10. After the test, compare the selected scenario with the expected scenario for each timestep.
11. Stop the test after 24 minutes.
12. Save the CSV file and screenshots.

CSV file

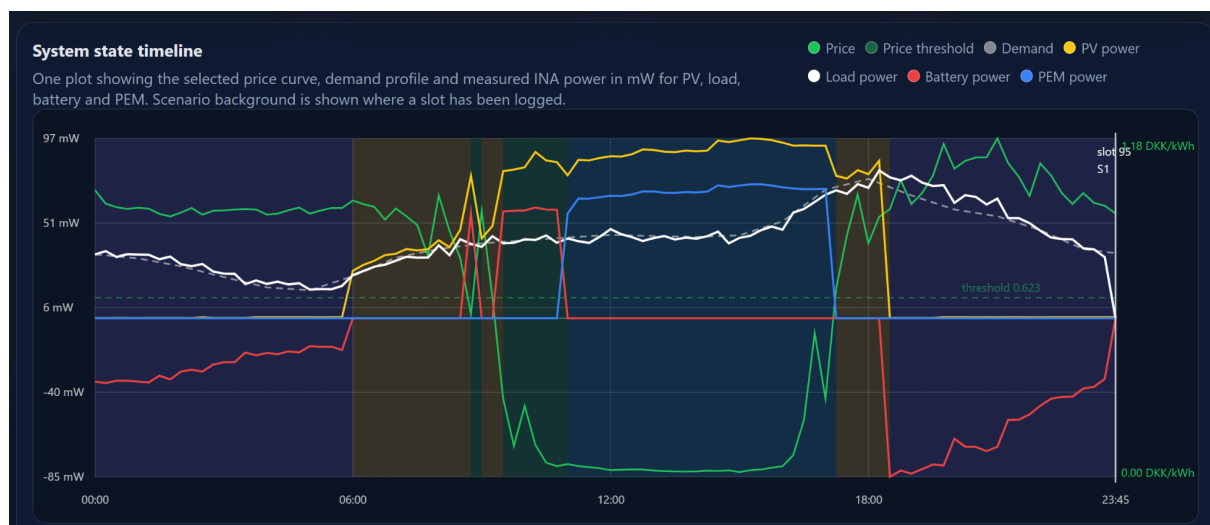
Item	Entry
CSV filename	/app/python/logs/ems_demo_20260512_151511.csv
Saved location	

Notes

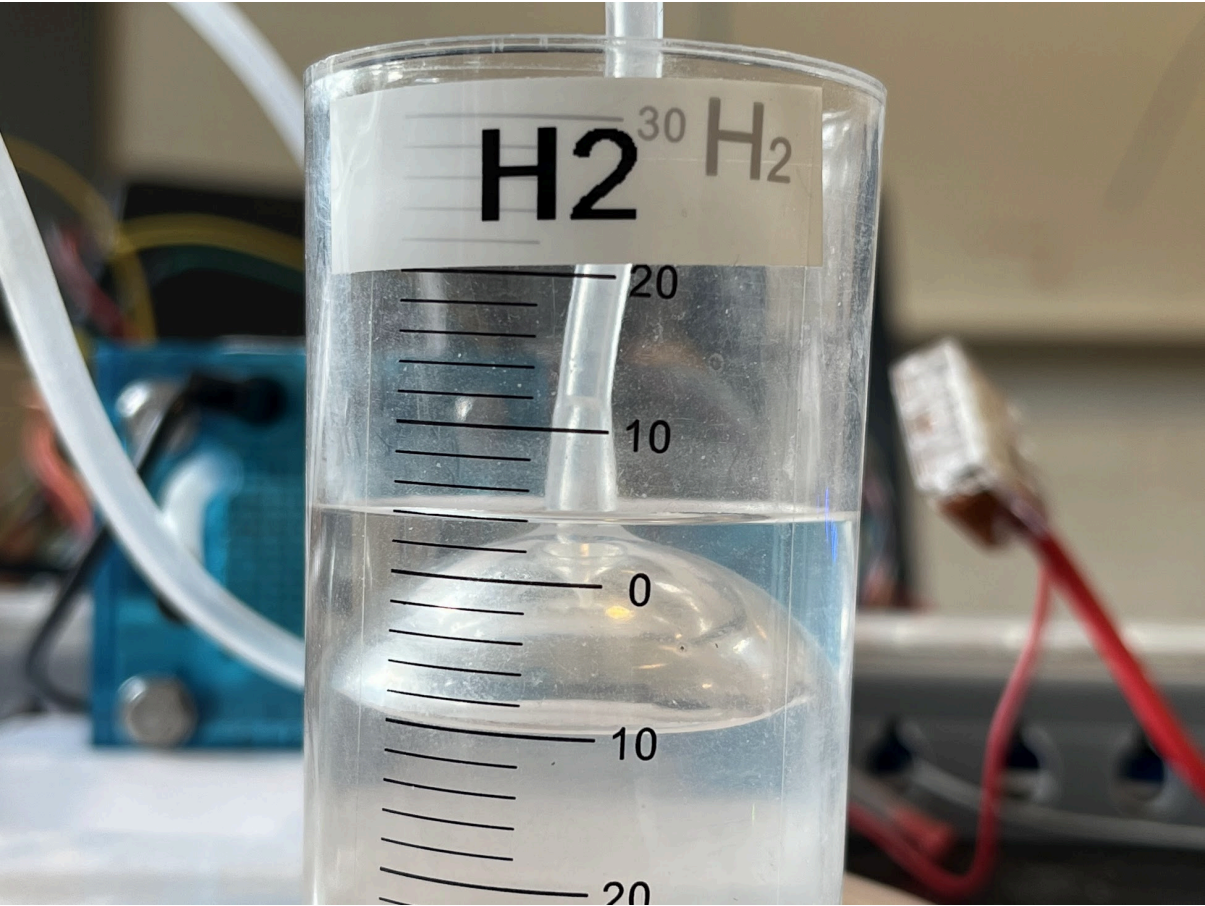
Time / timestep	Observation
00:00	S5 Battery to Load, Because of High Prices
06:00	S4 PV to Load, after lamp is turned on
08:45	S2 Grid to Load, PV to Batt. Single slot with low price
09:00	S4 PV to Load, after prices go to High again
09:30	S2 Grid to Load, PV to Batt. Prices Low
11:15	S3 Grid to Load, PV to PEM, Prices Low, batt charged
17:15	S4 PV to Load, Prices go high again
18:30	S5 Battery to Load, Because of High Prices, and PV no longer available

Result summary

Question	Answer
Did the EMS follow the expected logic throughout the test?	Yes
Which scenarios occurred during the test?	S2, S3, S4, S5
Did scenario changes match price transitions?	Yes, even if only a single slot
Did scenario changes match PV availability changes?	Yes
Did Battery SoC behave as expected?	Yes
Did PEM SoC behave as expected?	Yes
Did PEM only operate when load was below 40 mW?	PEM was never used.
Were there any unexpected scenario changes?	No
Overall result	Pass



Storage states		INA sensor measurements	
Current battery voltage	3.689 V	PV voltage	3.185 V
Initial SoC lookup voltage	3.692 V	Battery voltage	3.689 V
Lookup-estimated real battery SoC	4.6 %	PEM voltage	1.295 V
Integrated real battery SoC	4.6 %	Load voltage	5.120 V
Integrated real battery charge	91.266 mAh / 2000.0 mAh	PV current	0.13 mA
Virtual battery SoC	77.0 %	Battery current	-0.03 mA
Virtual battery capacity	10.00 mAh	PEM current	-0.04 mA
Virtual battery charge	7.705 mAh	Load current	-0.01 mA
Virtual battery state	medium	PV power	0.40 mW
Hydrogen volume	13.41 mL	Battery power	-0.09 mW
Usable hydrogen	5.81 mL	PEM power	-0.06 mW
Hydrogen estimation	72.6 %	Load power	-0.04 mW
Hydrogen mode	idle	Load trigger	LOW
		Arduino mode	S1 Grid -> Load



Hydrogen Niveau efter test (72.6% = 5.81mL) formentlig målt forkert ved start